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Balloon for expanding an orifice

Abstract

A balloon is disclosed for expanding an orifice. The balloon comprises a hollow body having a head. The body is closed at a distal end and open at a proximal end. The head extends from the distal end. The balloon also comprises a hollow neck positioned between the open proximal end and the distal end of the body. In the balloon, a wall thickness of the neck is greater than a wall thickness of the head. In a variation, the balloon can be configured such that, when the balloon is inflated, the head expands in a radial direction greater than in the longitudinal direction and a diameter of the body remains approximately constant during inflation. Also disclosed is a system for expanding an orifice. The system includes such a balloon and an inflation device for inflating the balloon. Also disclosed is a kit comprising the system and a container. Further, disclosed is a mould and a method for forming the balloon.

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Background/Summary

TECHNICAL FIELD

(1) This disclosure relates generally to balloons that are used to expand an orifice, such as a foreskin of a male subject.

BACKGROUND

(2) Phimosis is a condition in which the male foreskin is unable to retract properly from the head of the penis (or glans) due to an unusually tight foreskin. Many males with the condition are born with it. Phimosis can also be caused by trauma, infection or autoimmune disease. Zipper trauma is a well-documented cause of phimosis, and Lichen sclerosis and Balanitis Xerotica Obliterans (BXO) are autoimmune diseases known to trigger phimosis.

(3) Over time phimosis can also be exacerbated by tearing of the opening of the foreskin, resulting in scarring. Phimosis can result in a higher likelihood of infection, a build-up of smeg (the white substance under the foreskin), ballooning of the foreskin during urination, pain during sex and a loss of sensation during sex. A more serious condition, paraphimosis, occurs when a tight foreskin is pulled back over the head of the penis and remains there, constricting the blood flow. This can rapidly lead to permanent damage and/or loss of the penis.

(4) Devices used to treat phimosis insert a balloon under the foreskin where the balloon is then gently inflated which causes the foreskin to stretch. This procedure is repeated over a course of a few days to a few months to treat phimosis. A problem with current balloons is that they do not always stay in the correct position under the foreskin and they do not always have reproducible inflation characteristics, which can make it difficult to provide consistent treatment for a patient.

(5) It is to be understood that a reference herein to the prior art does not constitute an admission that the publication forms a part of the common general knowledge in the art, in Australia or any other country.

SUMMARY

(6) The disclosure provides in a first aspect a balloon for expanding an orifice. The balloon comprises a hollow body having a head, the body being closed at a distal end and open at a proximal end, the head extending from the distal end. The balloon also comprises a hollow neck

positioned between the open proximal end and the distal end of the body. A wall thickness of the neck is greater than a wall thickness of the head.

(7) An advantage of having a wall thickness of the neck being greater than a wall thickness of the head is that the neck may not expand, or there may be substantially minimised expansion of the neck, when the balloon is inflated. This may help to ensure that the expansion of the head during inflation is approximately uniform, which can help to ensure an even pressure is applied to the orifice being expanded by the balloon. For example, when treating phimosis, the approximately uniform expansion of the balloon may apply even pressure to the foreskin. Approximate uniform expansion of the head of the balloon may also help to ensure that the balloon does not slip out of the orifice in use.

(8) In an embodiment a wall thickness of the head at the distal end may be greater than the wall thickness of a remainder of the head. Having a wall thickness of the head at the distal end (e.g. a tip region/end of the head) being greater than the wall thickness of the remainder of the head may allow the balloon to expand in a radial direction more than a longitudinal direction extending from the proximal end to the distal end of the body during inflation. In some embodiments, the head may only expand in a radial direction during inflation i.e. the head of the balloon does not expand in the longitudinal direction. Limiting the expansion of the head to a radial direction may help to ensure that the balloon only exerts a radial force onto the orifice rather than a force in a longitudinal direction. For tissue such as the foreskin, minimising expansion in the longitudinal direction may help to prevent the balloon from being ejected out from under the foreskin during inflation.

(9) In an embodiment, a first end of the neck may extend from the proximal end of the body to a second remote end of the neck. The first end of the neck may comprise a peripheral lip that protrudes outwardly therefrom. The neck may comprise a waist intermediate its first and second ends. A wall thickness of the waist may be less than a wall thickness of a remainder of the neck. The waist may be formed as a progressive narrowing of the wall thickness of a section of the neck. Having a waist may help to give the body some flexibility to allow the body to be removed from a mould during manufacture. The waist may also help to reduce the amount of material required to form the balloon.

(10) An external diameter of the neck may be substantially constant for a length of the neck. The external diameter of the neck may range from about 5 mm to about 15 mm, such as about 10 mm. A portion of the neck may have an external diameter that is substantially constant, and another portion of the neck may have an external diameter that is not constant. The head may have a diameter ranging from about 6 mm to about 18 mm in a deflated state. The opening at the proximal end may have a diameter ranging from about 5 mm to about 10 mm. In an embodiment, the diameter of the opening is about 8 mm. The proximal end may be configured to fit onto an inflation device.

(11) The disclosure provides in a second aspect a balloon for expanding an orifice. The balloon comprises a hollow body having a head, the body being closed at a distal end and open at a proximal end with a longitudinal axis extending therebetween. The head extends from the distal end. The balloon is configured such that when the balloon is inflated the head expands in a radial direction greater than in the longitudinal direction and a diameter of the body remains approximately constant during inflation.

(12) A length of the balloon extending along the longitudinal direction may remain approximately constant during inflation. The body may define a neck that is positioned between the head and the proximal end. A diameter of the neck may remain substantially unchanged during inflation. The balloon of the second aspect may be otherwise as defined for the first aspect.

(13) In an embodiment of the disclosure, the balloon may be formed from silicone. The body and head may be unitary with one another. The balloon may be self-supporting such that in a deflated state the balloon does not collapse in on itself. Having a self-supporting balloon may assist a user inserting the balloon into an orifice. For example, when the orifice is an ear, a self-supporting

balloon may allow a user to insert the balloon into the ear irrespective of the patient's orientation, and when the orifice is the space between the foreskin and the head of a penis the self-supporting balloon may help with placement of the balloon under the foreskin.

(14) The disclosure also provides in another aspect a system for expanding an orifice, comprising: a balloon as set forth above, and an inflation device for inflating the balloon.

(15) The system may further comprise a flow regulator to regulate the flow of fluid into and out of the balloon. The flow regulator may be used to maintain the balloon in an inflated state. The system may further comprise a connector used to connect the balloon to the inflation device. The connector may function to clamp the proximal end to the balloon to the flow regulator.

(16) The disclosure also provides in another aspect a kit comprising a system as set forth above, and a container for housing the system. The container may be a box. The box may be cardboard and/or plastic. The kit may include instructions for operating the system.

(17) The disclosure also provides in another aspect a mould for forming a balloon that is used to expand an orifice. The balloon comprises a hollow body having a head, the body being closed at a distal end and open at a proximal end, and a hollow neck extending from the open proximal end of the body. The mould comprises a spigot having a first region for forming the body and a second region for forming the head. The first region has a first diameter and the second region has a second diameter, and the first diameter is less than the second diameter.

(18) The spigot may further comprise a third region for forming a waist in the neck. The third region may have a third diameter that is less than the first diameter. The second region may comprise a recess that forms a part of the proximal end of the body. The mould may further comprise a female section for receiving the spigot. The female section may have a bore that has an internal diameter that is approximately constant. The balloon may be as set forth above. The spigot may be formed from two or more components that are connectable with one another to form the spigot.

(19) Also disclosed is a method of forming a balloon, the balloon being used to expand an orifice. The method comprises: providing the mould as set forth above; applying a balloon precursor to the mould; and curing the balloon precursor to form the balloon.

(20) The balloon precursor may be a silicone precursor material that is cured to form a silicone-based balloon. The method may further comprise a post-curing step to further cure the balloon. The balloon may be as set forth above.

Description

BRIEF DESCRIPTION OF FIGURES

(1) Embodiments will now be described by way of example only with reference to the accompanying non-limiting Figures, in which:

(2) FIG. 1 is a side view of an embodiment of a balloon of the disclosure in a deflated state when mounted on a spigot.

(3) FIG. 2 is a side view of the balloon of FIG. 1 in an inflated state.

(4) FIG. 3 shows a partial cross-section of an embodiment of a balloon.

(5) FIG. 4 shows a partial cross-section of another embodiment of a balloon.

(6) FIG. 5 shows a cross-section of another embodiment of a balloon

(7) FIG. 6 shows a side view of an embodiment of a balloon of the disclosure in a deflated state when mounted on a spigot.

(8) FIG. 7 shows a cross-section of a mould that is used to form an embodiment of a balloon.

(9) FIG. 8 shows a male mould component that is used to form an embodiment of a balloon.

DETAILED DESCRIPTION OF EMBODIMENTS

(10) Disclosed is a balloon for expanding an orifice. Referring to FIGS. 1 and 2, the balloon 10 has

a hollow body in the form of hollow cylinder **16**. The cylinder **16** has a proximal (first) end in the form of a base **12**. Extending from the base **12** is a neck **22**. The neck **22** is integrally formed with the cylinder **16**. As best shown in FIG. 3, the cylinder **16** has an inner face **34** and an outer face **32**. A diameter of the outer face **32** remains approximately constant from the base **12** to the neck **22**. Extending from the neck **22** is a head **18**. Terminating at a distal end (e.g. tip) of the balloon **10** is a tip region in the form of end **20**. The neck **22** has a first end near (proximate) the base **12** and a second end remote (distal) from the base **12**. Put simply, each end of the neck **22** is axially spaced apart from one another.

(11) A lip in the form of resistance band **26** is positioned at the base **12** on the outer face **32**. Resistance band **26** acts to apply a compressive force when the base **12** is sleeved over a fitting that is associated with an inflation device to prevent leakage of fluid when the balloon **10** is inflated with the fluid. In FIGS. 1 and 2 the fitting is in the form of spigot **14**. An outer surface of the spigot **14** is in contact with the inner face **34** when the base **12** is sleeved over the spigot **14**, and the resistance band **26** acts to compress the inner face **34** onto the outer surface of the spigot **14**. A connector in the form of auxiliary ring **28** is also fitted over the cylinder **16** near the base **12** to compress the cylinder **16** onto the spigot **14**. The spigot **14** can form part of a flow regulator. An auxiliary ring **28** acts as a clamp, such as a sable clamp, to apply a radially compressive force to the cylinder **16**. The auxiliary ring **28** helps to ensure the base **12** of the balloon **10** remains attached to the spigot **14**. However, the auxiliary ring **28** is not required in all embodiments, see for example FIG. 6 where the balloon **10** is mounted on the spigot without the auxiliary ring. Additionally, the resistance band **26** is not required in all embodiments.

(12) The cylinder **16** has a waist positioned between the base **12** and the neck **22**. In the embodiment shown in FIGS. 1 to 3, the waist is in the form of a channel or groove **24** that extends circumferentially around the inner face **34**. The groove **24** has a depth that extends from the inner face **34** towards the outer face **32**. The groove **24** is formed as a progressive narrowing of the wall thickness of the cylinder **16**.

(13) In the embodiment depicted in FIGS. 1 to 3, a wall thickness of the cylinder **16** near the base **12** is approximately similar to a wall thickness of the neck **22**. In some embodiments the neck **22** has a wall thickness that is smaller or larger than the wall thickness of the cylinder near the base **12**. Regardless of the wall thickness of the neck **22** and cylinder **16**, a wall thickness of the groove **24**, as defined by the narrowest thickness between the inner face at location **34a** and outer face at location **32a**, is less than the wall thickness of the base **12**. The wall thickness of the groove **24** is also less than the wall thickness of the neck **22**. The narrowest thickness between the inner face **34a** and outer face **32a** of the groove **24** is generally located at the maximum groove depth, for example at location **34a**. The presence of groove **24** means that the neck **22** is defined by a circumferential projection that has a semi-hemispherical or curved cross-section that extends radially inwards. However, in embodiments where the groove **24** is positioned along the cylinder **16** at a location that is towards the base **12** more than the neck **22**, the neck **22** may simply be defined by the end region of the cylinder **16**. A wall thickness of the neck **22** is defined by the greatest distance between the inner face **34** and outer face **32**. A longitudinal axis of the balloon **10** is represented by dashed line **36**.

(14) The groove **24** is not required in all embodiments, and as shown in FIG. 4 the wall thickness of the cylinder **16** is constant from the base **12** to the neck **22**.

(15) In the embodiments shown in FIGS. 1-4 the head **18** has a bulbous (mushroom) shape. However, in an embodiment the head **18** is cylindrical and extends axially from the neck **22**, as shown in FIG. 5.

(16) The thickness of a sidewall portion **19** of the head **18** is less than the wall thickness at the neck **22**. The end **20** of the head **18** has a wall thickness that is greater than the sidewall portion **19**. The end **20** has a wall thickness that is generally less than the wall thickness of the neck **22**, although in some embodiments the end **20** may have the same wall thickness as the neck **22**. The transition

from the neck **22** to the head **18** is provided as a progressive narrowing of the wall thickness of the cylinder **16** from the neck **22** to sidewall portion **19**.

(17) Specific dimensions of an embodiment of the balloon **10** in a deflated state (or uninflated state or non-expanded state) will now be described with reference to FIG. 5. It should be appreciated that the dimensions described in relation to the embodiment of FIG. 5 are exemplary only and are not intended to limit the dimensions of the balloon as set forth in this disclosure.

(18) In the balloon shown in FIG. 5, the base **12** has a wall thickness A' that in one embodiment ranges from about 1.0 mm-2.0 mm, such as about 1.20 mm. In a specific embodiment, A' is 1.22 mm. The cylinder **16** has a wall thickness extending from the inner face **34a** and outer face **32a** ranging from about 0.3 mm-0.7 mm, such as about 0.50 mm. In a specific embodiment, a wall thickness extending from the inner face **34a** to outer face **32a** is 0.52 mm. The neck **22** has a wall thickness B' ranging from about 1.0 mm-2.0 mm, such as about 1.20 mm. In a specific embodiment B' is 1.21 mm. The sidewall portion **19** of the head **18** has a wall thickness C' ranging from about 0.2 mm-0.6 mm, such as about 0.3 mm-0.4 mm. In a specific embodiment C' is 0.35 mm. The end **20** has a wall thickness D' ranging from about 0.7 mm-1.1 mm, such as about 0.90 mm. In a specific embodiment D' is 0.89 mm.

(19) Still referring to FIG. 5, a length I' from the base **12** to an intermediate point **16'** of the cylinder **16** extending along an axial direction **36** of the balloon **10** ranges from about 10 mm-15 mm. In a specific embodiment the length I' is about 13.5 mm. A length H' from the intermediate point **16'** to a distal end **16'** of the cylinder **16** extending along the axial direction **36** of the balloon **10** ranges from about 10 mm-15 mm. In a specific embodiment length H' is about 12 mm. The neck **22** has a length G' extending along the axial direction **36** of the balloon **10** that ranges from about 1.0 mm-3.0 mm. In a specific embodiment length G' is about 2.0 mm. A length E' of the head **18** extending along the axial direction **36** from the neck **22** to the end **20** ranges from about 10 mm-15 mm. In a specific embodiment length E' is about 12.50 mm, such as 12.48 mm.

(20) Still referring to FIG. 5, in an embodiment a radius J' from the axis **36** to inner wall **34** of the base **12** ranges from about 3.0 mm-4.5 mm. In a specific embodiment J' is 3.80 mm. In a specific embodiment, a radius K' from the longitudinal axis **36** to outer wall **32** of the base **12** is about 5 mm. It should be appreciated that the radii of the various features from the axis **36** to the inner and outer walls of the respective feature will be dependent upon a wall thickness of the respective feature and overall diameter of the balloon **10**. The dimensions of the balloon **10** described with reference to FIG. 5 are applicable to the embodiments of the balloon **10** shown in FIGS. 1-4.

(21) In the embodiments shown in the Figures the cylinder **16**, base **12**, resistance band **26**, groove **24**, neck **22**, head **18** and end **20** are unitary with one another. However, in some embodiments the cylinder **16**, base **12**, resistance band **26**, groove **24**, neck **22**, head **18** and end **20** are not unitary with one another.

(22) When the balloon **10** is inflated, such as when the balloon **10** is used to stretch the foreskin of a patient, a fluid is passed through spigot **14**, through the base **12** and into an interior space of the balloon **10**. The fluid is generally air but may optionally be a liquid. Because the head **18** has the thinnest wall thickness out of the cylinder **16**, base **12** and neck **22**, the head has the greatest ability to stretch (e.g. elasticity) which means that the head **18** is the first region of the balloon **10** to expand. Further, the wall thickness of the neck **22** is chosen so that the neck **22** does not expand, or only minimally expands, when the balloon **10** is inflated. The wall thickness of the cylinder **16** (e.g. groove **24** and base **12**) is also selected so that during inflation of the balloon **10** the cylinder **16** does not expand, that is, a diameter of the cylinder **16** remains substantially unchanged. See, for example, FIGS. 1 and 2, where in the inflated state (FIG. 2), a diameter of the outer face **32** of the cylinder **16** including neck **22** remains substantially unchanged compared to the neck **22** in a deflated state (FIG. 1). An advantage of the balloon **10** is that the head region **18** preferentially expands compared to the cylinder **16** (i.e. body), and this means that for small and delicate orifices the head **18** is less likely to slide out from the orifice. Put another way, the wall thickness of the

neck **22** and cylinder **16** relative the wall thickness of the head **18** helps to keep fluid at the head (i.e. the distal end) of the balloon **10** in an inflated state.

(23) For example, for current balloons used to dilate and stretch foreskin, there is no preferential expansion built into the balloon and the shaft and head expand proportionally with one another. A result of this is that for foreskins that are sufficiently tight, the head of the balloon is not able to expand and instead the shaft expands in preference to the head. This results in the shaft applying pressure on an outside of the foreskin which causes the head to withdraw from under the foreskin. This makes it difficult to reproducibly treat patients. Since the cylinder **16**, including neck **22**, does not expand during inflation, the cylinder **16** (i.e. body) is unlikely to apply a withdrawing pressure to the foreskin which helps to maintain the head **18** of the balloon **10** under the foreskin. Therefore, an embodiment of the balloon **10** helps to reduce the occurrence of the head withdrawing from an orifice in use.

(24) It should be appreciated that the neck **22** and cylinder **16** may expand slightly during inflation, thereby increasing their respective diameters, but such expansion is minimal i.e. a diameter of the neck **22** and cylinder **16** may expand by <10%. In an embodiment, the head **18** expands by up to 100% of its deflated state. For example, in embodiments where the head **18** in a deflated state has a diameter of about 12 mm and the neck **22** has a diameter of about 10 mm, the head **18** in the inflated state may expand to have a diameter of up to 24 mm, such as 18 mm, and a diameter of the neck **22** may remain relatively unchanged such as 10.5 mm. In an embodiment the head **18** has an outer diameter in a deflated state ranging from about 6 mm to about 18 mm. In an embodiment, the cylinder **16** has an outer diameter in a deflated state ranging from about 8 mm to about 12 mm. In some embodiments the outer diameter of the cylinder **16** in the deflated state remains unchanged regardless of the diameter of the head **18** in the deflated state. In some embodiments the neck **22** expands relative the base **12**, and the head **18** expands relative the neck **22**.

(25) A longitudinal length of the balloon **10** extends in a direction extending from the base **12** to the end **20**, as depicted as dashed line **36**. In an embodiment, the head **18** occupies approximately 20%-30% of the length of the balloon **10**. In an embodiment the neck **22** is positioned approximately 60%-75% of the distance from the base **12** to the end **20**. In an embodiment, a length of the balloon **10** ranges from about 30 mm to about 50 mm. In an embodiment, the head extends from about 10 mm to about 25 mm from the end **20** towards the base **12**. In an embodiment, a distance from the base **12** to the neck **22** ranges from about 20 mm to about 30 mm.

(26) In the embodiments shown in the Figures the end **20** of the head **18** has a wall thickness that is greater than the sidewall portion **19**. Similar to the neck **22**, the end **20** functions to limit the expansion of the balloon **10** in a region near the end **20**. Limiting the expansion of the end **20** helps to ensure that the head **18** expands in a radial direction rather than in a longitudinal direction of the balloon **10** (e.g. in a direction along line **36** extending away from the base **12**). In some embodiments, an overall length of the balloon **10** remains substantially unchanged (i.e. <15% change) between a deflated and inflated state.

(27) The balloon **10** is made from a resiliently deformable material. In an embodiment, the balloon **10** is formed from a material that allows the balloon **10** to be self-supporting such that in a deflated state the balloon **10** does not collapse in on itself, as depicted in FIG. **1**. In an embodiment the balloon is formed from a silicone-based material. For example, the balloon may be formed from a silicone rubber made from a SILPURAN® 6000/40 liquid silicone rubber precursor. In an embodiment the balloon is formed from a polymeric material such as rubber.

(28) The balloon **10** is used to dilate and stretch an orifice such as that formed by a foreskin. To stretch a foreskin, the head **18** of the balloon **18** is first inserted underneath the foreskin. Generally, the end **20** is positioned to be at or near the head of the penis. In an embodiment, an insertion implement, such as a rod, is used to push the head **18** under the foreskin. The insertion implement can be inserted into an interior of the balloon **10** to help push the head **18** underneath the foreskin. As the balloon **10** can be self-supporting, this can help with inserting the balloon **10** and

maintaining the balloon **10** under the foreskin.

(29) The balloon **10** is inflated once the balloon **10** positioned under the foreskin. To inflate the balloon **10** an inflation device is connected to the base **12** and a fluid is passed into the balloon **10**. The spigot **14** forms part of an inflation device. An example of an inflation device includes a syringe or a squeeze bulb. In some forms the inflation device has a flow regulator, such as a tap or valve, that can be used to regulate a flow of fluid into and out of the balloon **10**. The valve may be a one-way valve. The flow regulator is operable to maintain the balloon **10** in an inflated (expanded) state. In the embodiments of FIGS. **1** and **2** the auxiliary ring **28** is placed over balloon **10** to secure the balloon **10** to the spigot **14**. However, the auxiliary ring **28** is not required in all embodiments, as shown in FIG. **6**.

(30) In some embodiments, a kit is provided which comprises the balloon **10** and a container that houses the balloon **10**. In some embodiments, a kit is provided which comprises the system that comprises the balloon **10**, and a container which houses the system. The container may be made from cardboard and/or plastic. The container may contain recesses to house the balloon **10** and/or system. For example, a container can contain a first recess to hold a plurality of balloons **10**, and a second recess to hold the system.

(31) Referring now to FIG. **7**, a mould **100** is shown that is used to form the balloon **10**. The mould **100** has a first component in the form of a female component **110**. Female component **110** has a cavity **120**. In an embodiment, the cavity **120** is defined by an elongate blind bore having an approximately constant diameter and a rounded base. An upper region of the cavity **120** is defined by cavity side wall **112**. Extending from the cavity side wall **112** is a curved wall **124**. An end of the cavity **120** is defined by a base wall **126**. The base wall **126** extends from the curved wall **124**. The cavity side wall **112**, curved wall **124** and base wall **126** are collectively termed the cavity wall. The cavity wall defines an outer surface of the balloon **10**. In an embodiment the female component **110** is formed from Calmax tool steel.

(32) Mould **100** has a second component in the form of a male component, which in FIG. **7** is shown as spigot **114**. Spigot **114** has a base **125** which extends from a base block **131**. Base block **131** engaged with the female mould **110** to limit movement of the spigot **114** into the cavity **120**. Extending from the base **125** is a first spigot region in the form of base region **127** for forming the base **12** and cylinder **16** of the balloon **10**. A second spigot region in the form of head region **118** extends from the base region **127**. The head region **118** is used to define the head **18** of the balloon **10**. A diameter of the base region **127** is less than a diameter of the head region **118**. Positioned between the base region **127** and head region **118** is an intermediate region **116** that is used to define the groove **24** of the balloon **10**. A diameter of the intermediate region **116** can be the same or less than the diameter of the head region **118**, but the diameter of the intermediate region **116** is greater than the diameter of the base region **127**.

(33) In the embodiment of FIG. **7**, the neck **22** is formed around a neck region **128** that is positioned between the intermediate region **116** and head region **118**. In embodiments of the mould **100** that do not have the intermediate region **116**, the neck **22** is defined by the transition from the base region **127** to the head region **118**. In an embodiment the spigot **114** is formed from 7000 Series aluminium.

(34) The dimensions of features of the balloon **10** described with reference to FIG. **5**, such as wall thickness, length, and diameter, are at least in part defined by the dimensions of the cavity **120**, base region **127**, head region **118**, intermediate region **116** and neck region **128**. Adjusting the dimensions of the cavity **120**, base region **127**, head region **118**, intermediate region **116** and/or neck region **128** spigot will adjust the dimensions of the respective feature of a balloon **10**. For example, decreasing a diameter of the base region **127** whilst maintaining a diameter of the cavity **120** will increase a wall thickness of the cylinder **16**.

(35) In an embodiment, the spigot **114** is formed from two or more components that are connectable with one another. Forming the spigot from two or more components can help in the

removal of a balloon from the mould. In an embodiment, the base region **127**, intermediate region **116** and optionally the neck region **128** is defined by a first spigot component and the head region **118** and optionally the neck region **128** is defined by a second spigot component that is connectable with the first spigot component. In an embodiment, the neck region is positioned on the first spigot component, the second spigot component. The neck region **128** may be defined when the first spigot component and the second spigot component are connected to one another.

(36) In use, the spigot **114** is inserted into the cavity **120**. An annulus **122** formed between the cavity wall and an outer surface of the spigot defines a volume that is filled with a balloon precursor material that is cured to form the balloon. In an embodiment, the balloon precursor material is SILPURAN® 6000/40 liquid silicone rubber precursor. Once the balloon **10** is formed, the spigot **114** is removed from the cavity **120** and then the balloon **10** is removed from the spigot **114**. In an embodiment, following the formation of the balloon **10**, the balloon is post-cured. Post curing can be at an elevated temperature, such as about 200° C. However, post-curing is not required in all embodiments. A suitable release agent is used to allow the balloon **10** to be released from the mould **100**.

(37) An embodiment of a spigot is shown in FIG. **8** (multiple spigots are shown in FIG. **8**). The spigot **150** has base region **127** and intermediate region **116**. The spigot **150** extends from a base block **131**. In FIG. **8**, the base block **131** has a plurality of spigots **150** located thereat. A female component that is used to form the mould **100** is provided with a respective number of cavities to accommodate each spigot of the plurality of spigots **150**.

(38) Although specific embodiments of the balloon **10** have been described with reference to stretching a foreskin, the balloon of the current disclosure is not limited to use for stretching and dilating foreskin and can be used for dilating and stretching other orifices such as an ear canal, nasal passage, sinus, anus and vagina.

(39) Further, the embodiments of the balloon described above have been designed for specific use for stretching a foreskin. However, for use with other orifices, other features of the balloon **10**, such as a portion of the cylinder **16**, may expand similarly or more than the head **18** in use. For example, in some embodiments, the balloon in an expanded shape may adopt a “dog bone” shape, with two expanded portions being axially spaced from one another.

(40) Having described the subject matter of the present disclosure in detail and by reference to specific embodiments thereof, it is noted that the various details disclosed herein should not be taken to imply that these details relate to elements that are essential components of the various embodiments described herein, even in cases where a particular element is illustrated in each of the drawings that accompany the present description. Further, it will be apparent that modifications and variations are possible without departing from the scope of the present disclosure, including, but not limited to, embodiments defined in the appended claims. More specifically, although some aspects of the present disclosure may be identified herein as preferred or particularly advantageous, it is contemplated that the present disclosure is not necessarily limited to these aspects.

(41) In the claims which follow and in the preceding description, except where the context requires otherwise due to express language or necessary implication, the word “comprise” or variations such as “comprises” or “comprising” is used in an inclusive sense, i.e. to specify the presence of the stated features but not to preclude the presence or addition of further features in various embodiments.

Claims

1. A balloon configured for expanding a foreskin, the balloon comprising: a hollow body having a head and a neck, the neck having a length and the body being closed at a distal end and open at a proximal end with a longitudinal axis extending therebetween, the body having an inner face and an outer face defining a wall thickness of the body, the head extending from the neck and

terminating at the closed distal end, the closed distal end defined by a transverse end wall being that portion of the head which extends across the distal end of the body in a direction transverse to the longitudinal axis, the head having a sidewall portion being that portion of the head which is directly connected to extend between each of the transverse end wall and the neck; wherein a wall thickness of the neck is greater than a wall thickness of the sidewall portion of the head, and a wall thickness of the transverse end wall is greater than a wall thickness of the sidewall portion of the head, such that when the balloon is inflated the head expands in a radial direction greater than in a direction of the longitudinal axis and a diameter of an outer face of the neck remains approximately constant over the length of the neck during inflation, to thereby form the balloon with a bulbous mushroom shape, the bulbous mushroom head having an outer diameter greater than an outer diameter of the neck, the outer diameter of the neck being substantially cylindrical.

2. The balloon as claimed in claim 1, wherein a length of the balloon extending along the direction of the longitudinal axis remains approximately constant during inflation.
 3. The balloon as claimed in claim 1, wherein a first end of the neck comprises a peripheral lip that protrudes outwardly therefrom.
 4. The balloon as claimed in claim 1, wherein the neck comprises a waist intermediate a first end and a second end of the neck.
 5. The balloon as claimed in claim 4, wherein a wall thickness of the waist is less than a wall thickness of a remainder of the neck.
 6. The balloon as claimed in claim 4, wherein the waist is formed as a progressive narrowing of a wall thickness of a section of the neck.
 7. The balloon as claimed in claim 1, wherein the external diameter of the neck ranges from about 5 mm to about 15 mm.
 8. The balloon as claimed in claim 1, wherein the head has a diameter within a range from about 6 mm to about 18 mm in a deflated state.
 9. The balloon as claimed in claim 1, wherein the opening at the proximal end has a diameter within a range from about 5 mm to about 10 mm.
 10. The balloon as claimed in claim 1, wherein the balloon is formed from silicone.
 11. The balloon as claimed in claim 10, wherein the balloon is formed from a liquid silicone rubber precursor.
 12. The balloon as claimed in claim 1, wherein the body and head are unitary with one another.
 13. The balloon as claimed in claim 1, being self-supporting such that in a deflated state the balloon does not collapse in on itself.
 14. The balloon as claimed in claim 9, wherein the opening at the proximal end has a diameter of 8 mm.
 15. The balloon as claimed in claim 1, wherein the wall thickness of the neck is within a range from 1.0 mm-2.0 mm, the wall thickness of the sidewall portion of the head is within a range from 0.2 mm-0.6 mm and the wall thickness of the transverse end wall is within a range from 0.7 mm-1.1 mm.
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